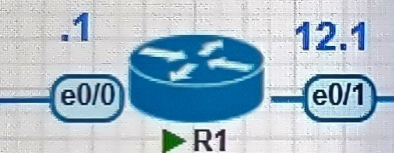
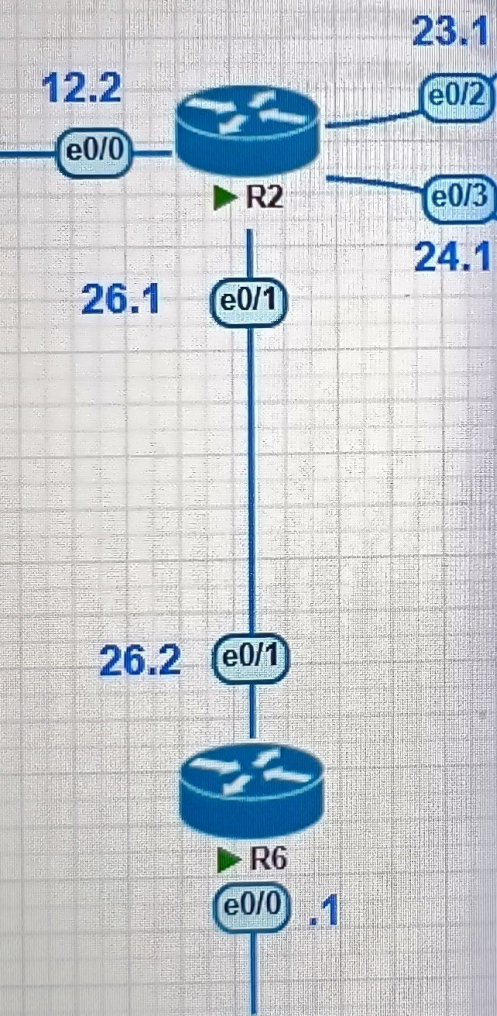


0,000 RS



```
wn
*Aug 24 06:57:15.465: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
2, changed state to down
*Aug 24 06:57:15.470: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
3, changed state to down
R1#
R1#
R1#CONF T
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int e0/0
R1(config-if)#ip add 192.50.20.0 255.255.255.0
Bad mask /24 for address 192.50.20.0
R1(config-if)#ip add 192.50.20.1 255.255.255.0
R1(config-if)#no sh
R1(config-if)#exit
R1(config)#
R1(config)#
R1(config)#int e0/1
R1(config-if)#ip add 12.0.0.1 255.0.0.0
R1(config-if)#no sh
R1(config-if)#exit
R1(config)#
R1(config)#
*Aug 24 07:02:04.094: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Confi
g Wizard)
R1(config)#
```

```
R2#conf t
```

```
Enter configuration commands, one per line. End with CNTL/Z.
```

```
R2(config)#int e0/0
```

```
R2(config-if)#ip add 12.0.0.2 255.0.0.0
```

```
R2(config-if)#no sh
```

```
R2(config-if)#exit
```

```
R2(config)#
```

```
R2(config)#int e0/1
```

```
R2(config-if)#ip add 26.0.0.1 255.0.0.0
```

```
R2(config-if)#no sh
```

```
R2(config-if)#exit
```

```
R2(config)#
```

```
R2(config)#int e0/2
```

```
R2(config-if)#ip add
```

```
% Incomplete command.
```

```
R2(config-if)#ip add 23.0.0.1 255.0.0.0
```

```
R2(config-if)#no sh
```

```
R2(config-if)#exit
```

```
R2(config)#
```

```
R2(config)#int e0/3
```

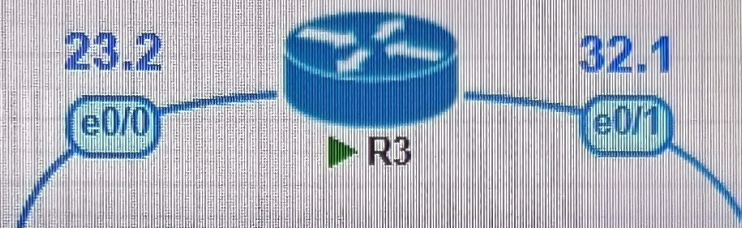
```
R2(config-if)#ip add 24.0.0.1 255.0.0.0
```

```
R2(config-if)#no sh
```

```
R2(config-if)#exit
```

```
R2(config)#
```

```
*Aug 24 07:04:13.491: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
```

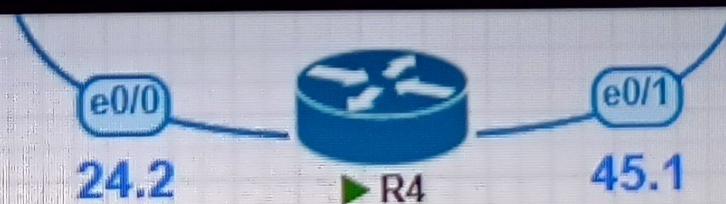



PuTTY (inactive)

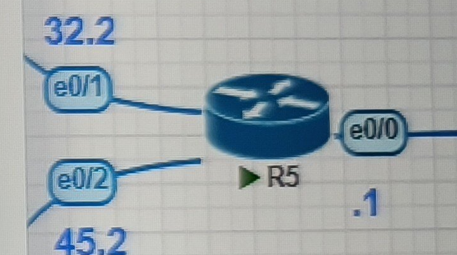
```
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#
R3(config)#int e0/0
R3(config-if)#ip add 23.0.0.2 255.0.0.0
R3(config-if)#no sh
R3(config-if)#exit
R3(config)#
R3(config)#int e0/1
R3(config-if)#ip add 32.0.0.1 255.0.0.0
R3(config-if)#no sh
R3(config-if)#exit
R3(config)#
R3(config)#
*Aug 24 07:03:59.820: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
```



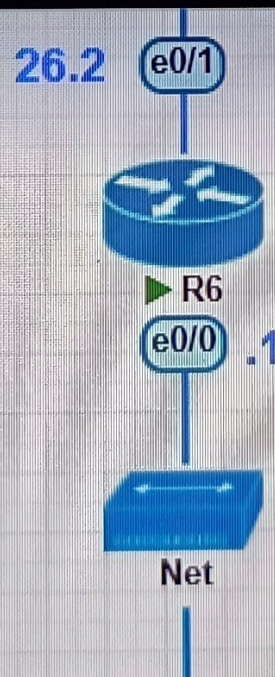
```
R4#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R4(config)#
R4(config)#int e0/0
R4(config-if)#ip add 24.0.0.2 255.0.0.0
R4(config-if)#no sh
R4(config-if)#exit
R4(config)#
R4(config)#int e0/1
R4(config-if)#ip add 45.0.0.1 255.0.0.0
R4(config-if)#no sh
R4(config-if)#exit
R4(config)#
*Aug 24 07:09:13.517: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R4(config)#
```

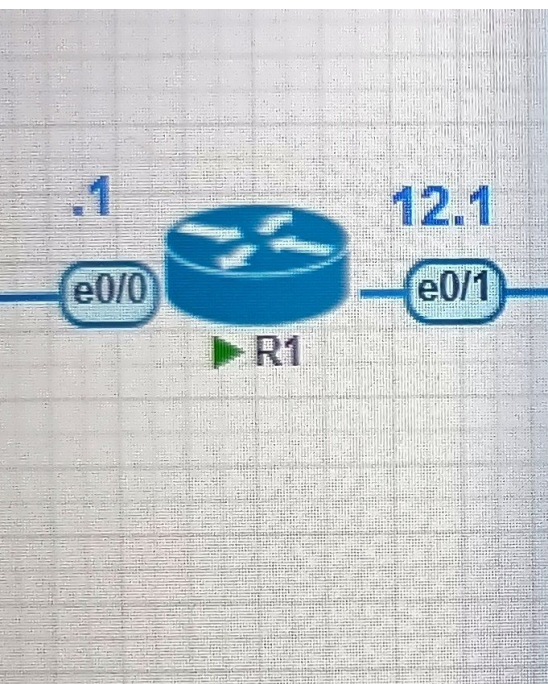



```
R5#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R5(config)#int e0/0
R5(config-if)#ip add
*Aug 24 07:05:51.399: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R5(config-if)#ip add 172.36.0.1 255.255.0.0
R5(config-if)#no sh
R5(config-if)#exit
R5(config)#
R5(config)#int e0/1
R5(config-if)#ip add 32.0.0.2 255.0.0.0
R5(config-if)#no sh
R5(config-if)#exit
R5(config)#
R5(config)#int e0/2
R5(config-if)#ip add 45.0.0.2 255.0.0.0
R5(config-if)#no sh
R5(config-if)#exit
R5(config)#
```




```
R6#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R6(config)#int e0/0
R6(config-if)#ip add 223.0.50.1 255.255.255.0
R6(config-if)#no sh
R6(config-if)#exit
R6(config)#
R6(config)#int e0/1
R6(config-if)#ip add 26.0.0.2 255.0.0.0
R6(config-if)#no sh
R6(config-if)#exit
R6(config)#
*Aug 24 07:10:33.103: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R6(config)#
```





```
R1(config)#ip route 23.0.0.0 255.0.0.0 12.0.0.2
R1(config)#ip route 24.0.0.0 255.0.0.0 12.0.0.2
R1(config)#ip route 26.0.0.0 255.0.0.0 12.0.0.2
R1(config)#ip route 32.0.0.0 255.0.0.0 12.0.0.2
R1(config)#ip route 45.0.0.0 255.0.0.0 12.0.0.2
R1(config)#ip route 172.36.0.0 255.255.0.0 12.0.0.2
R1(config)#ip route 223.0.50.0 255.255.255.0 12.0.0.2
R1(config)#
R1(config)#
```


R1

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks

C 12.0.0.0/8 is directly connected, Ethernet0/1

L 12.0.0.1/32 is directly connected, Ethernet0/1

S 23.0.0.0/8 [1/0] via 12.0.0.2

S 24.0.0.0/8 [1/0] via 12.0.0.2

S 26.0.0.0/8 [1/0] via 12.0.0.2

S 32.0.0.0/8 [1/0] via 12.0.0.2

S 45.0.0.0/8 [1/0] via 12.0.0.2

S 172.36.0.0/16 [1/0] via 12.0.0.2

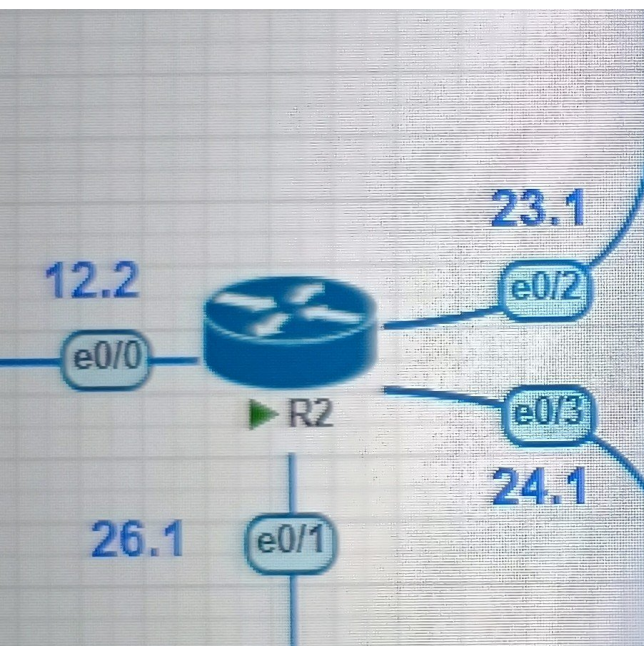
192.50.20.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.50.20.0/24 is directly connected, Ethernet0/0

L 192.50.20.1/32 is directly connected, Ethernet0/0

S 223.0.50.0/24 [1/0] via 12.0.0.2

R1#



```
R2(config)#  
R2(config)#ip route 192.50.20.0 255.255.255.0 12.0.0.1  
R2(config)#ip route 223.0.50.0 255.255.255.0 26.0.0.2  
R2(config)#ip route 32.0.0.0 255.0.0.0 23.0.0.2  
R2(config)#ip route 45.0.0.0 255.0.0.0 24.0.0.2  
R2(config)#ip route 172.36.0.0 255.255.0.0 23.0.0.2  
R2(config)#
```


o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from Pfr

Gateway of last resort is not set

12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks

12.0.0.0/8 is directly connected, Ethernet0/0

12.0.0.2/32 is directly connected, Ethernet0/0

23.0.0.0/8 is variably subnetted, 2 subnets, 2 masks

23.0.0.0/8 is directly connected, Ethernet0/2

23.0.0.1/32 is directly connected, Ethernet0/2

24.0.0.0/8 is variably subnetted, 2 subnets, 2 masks

24.0.0.0/8 is directly connected, Ethernet0/3

24.0.0.1/32 is directly connected, Ethernet0/3

26.0.0.0/8 is variably subnetted, 2 subnets, 2 masks

26.0.0.0/8 is directly connected, Ethernet0/1

26.0.0.1/32 is directly connected, Ethernet0/1

32.0.0.0/8 [1/0] via 23.0.0.2

45.0.0.0/8 [1/0] via 24.0.0.2

172.36.0.0/16 [1/0] via 23.0.0.2

192.50.20.0/24 [1/0] via 12.0.0.1

223.0.50.0/24 [1/0] via 26.0.0.2



PuTTY (inactive)

```
R3(config)#ip route 192.50.20.0 255.255.255.0 23.0.0.1
R3(config)#ip route 12.0.0.0 255.0.0.0 23.0.0.1
R3(config)#ip route 172.36.0.0 255.255.0.0 32.0.0.2
R3(config)#ip route 24.0.0.0 255.0.0.0 23.0.0.1
R3(config)#ip route 26.0.0.0 255.0.0.0 23.0.0.1
R3(config)#ip route 45.0.0.0 255.0.0.0 23.0.0.1
R3(config)#ip route 223.0.50.0 255.255.255.0 23.0.0.1
R3(config)#
```


N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from Pfr

Gateway of last resort is not set

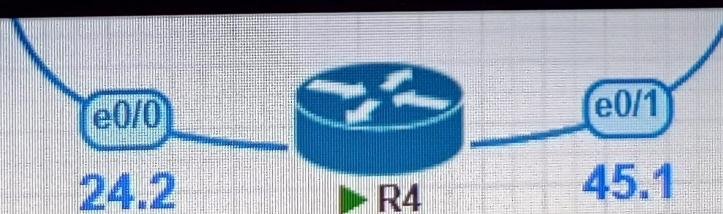
```
S    12.0.0.0/8 [1/0] via 23.0.0.1
    23.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    23.0.0.0/8 is directly connected, Ethernet0/0
L    23.0.0.2/32 is directly connected, Ethernet0/0
S    24.0.0.0/8 [1/0] via 23.0.0.1
S    26.0.0.0/8 [1/0] via 23.0.0.1
    32.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    32.0.0.0/8 is directly connected, Ethernet0/1
L    32.0.0.1/32 is directly connected, Ethernet0/1
S    45.0.0.0/8 [1/0] via 23.0.0.1
S    172.36.0.0/16 [1/0] via 32.0.0.2
S    192.50.20.0/24 [1/0] via 23.0.0.1
S    223.0.50.0/24 [1/0] via 23.0.0.1
```

R3#

R3#

R4

```
R4(config)#ip route 172.36.0.0 255.255.0.0 45.0.0.2
R4(config)#ip route 192.50.200.0 255.255.255.0 24.0.0.1
R4(config)#ip route 223.0.50.0 255.255.255.0 24.0.0.1
R4(config)#ip route 12.0.0.0 255.0.0.0 24.0.0.1
R4(config)#ip route 23.0.0.0 255.0.0.0 24.0.0.1
R4(config)#ip route 32.0.0.0 255.0.0.0 45.0.0.2
R4(config)#ip route 26.0.0.0 255.0.0.0 24.0.0.1
R4(config)#
```



E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

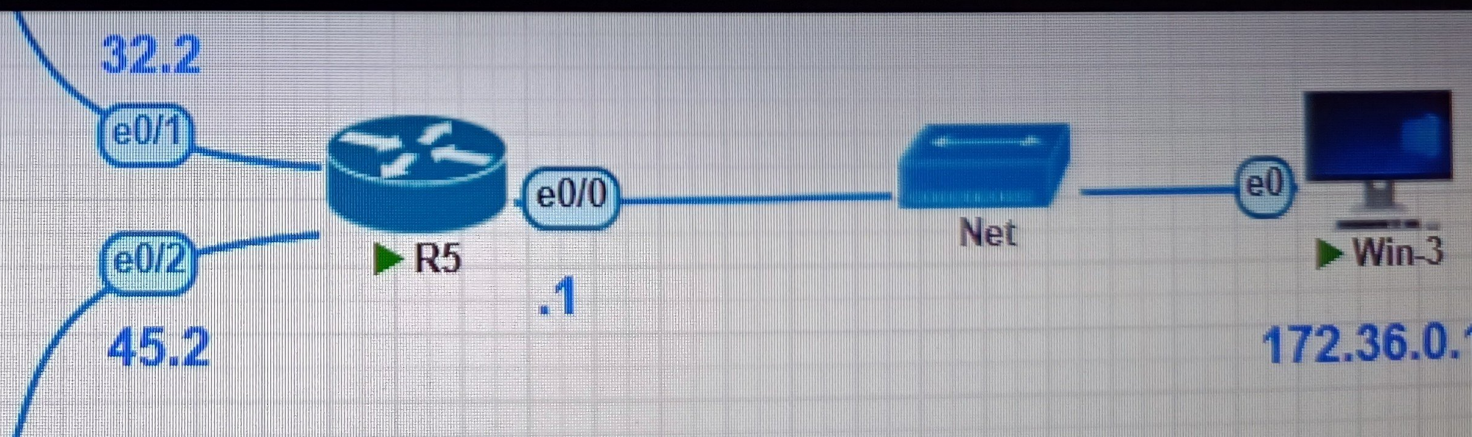
```
S 12.0.0.0/8 [1/0] via 24.0.0.1
S 23.0.0.0/8 [1/0] via 24.0.0.1
C 24.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L   24.0.0.0/8 is directly connected, Ethernet0/0
L   24.0.0.2/32 is directly connected, Ethernet0/0
S 26.0.0.0/8 [1/0] via 24.0.0.1
S 32.0.0.0/8 [1/0] via 45.0.0.2
C 45.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L   45.0.0.0/8 is directly connected, Ethernet0/1
L   45.0.0.1/32 is directly connected, Ethernet0/1
S 172.36.0.0/16 [1/0] via 45.0.0.2
S 192.50.200.0/24 [1/0] via 24.0.0.1
S 223.0.50.0/24 [1/0] via 24.0.0.1
```

R4#

R4#

R5

```
R5(config)#ip route 23.0.0.0 255.0.0.0 32.0.0.1
R5(config)#ip route 24.0.0.0 255.0.0.0 45.0.0.1
R5(config)#ip route 26.0.0.0 255.0.0.0 45.0.0.1
R5(config)#ip route 12.0.0.0 255.0.0.0 32.0.0.1
R5(config)#ip route 192.50.20.0 255.255.255.0 32.0.0.1
R5(config)#ip route 223.0.50.0 255.255.255.0 45.0.0.1
R5(config)#
R5(config)#exit
R5#s
```



ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PIR

Gateway of last resort is not set

```
S 12.0.0.0/8 [1/0] via 32.0.0.1
S 23.0.0.0/8 [1/0] via 32.0.0.1
S 24.0.0.0/8 [1/0] via 45.0.0.1
S 26.0.0.0/8 [1/0] via 45.0.0.1
32.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C   32.0.0.0/8 is directly connected, Ethernet0/1
L   32.0.0.2/32 is directly connected, Ethernet0/1
45.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C   45.0.0.0/8 is directly connected, Ethernet0/2
L   45.0.0.2/32 is directly connected, Ethernet0/2
172.36.0.0/16 is variably subnetted, 2 subnets, 2 masks
C   172.36.0.0/16 is directly connected, Ethernet0/0
L   172.36.0.1/32 is directly connected, Ethernet0/0
S 192.50.20.0/24 [1/0] via 32.0.0.1
S 223.0.50.0/24 [1/0] via 45.0.0.1
```

R5#

R5#



R6



```
R6(config)#ip route 192.50.20.0 255.255.255.0 26.0.0.1
R6(config)#ip route 172.36.0.0 255.255.0.0 26.0.0.1
R6(config)#ip route 12.0.0.0 255.0.0.0 26.0.0.1
R6(config)#ip route 23.0.0.0 255.0.0.0 26.0.0.1
R6(config)#ip route 24.0.0.0 255.0.0.0 26.0.0.1
R6(config)#ip route 32.0.0.0 255.0.0.0 26.0.0.1
R6(config)#ip route 45.0.0.0 255.0.0.0 26.0.0.1
R6(config)#
```

26.2

e0/1



R6

e0/0

.1

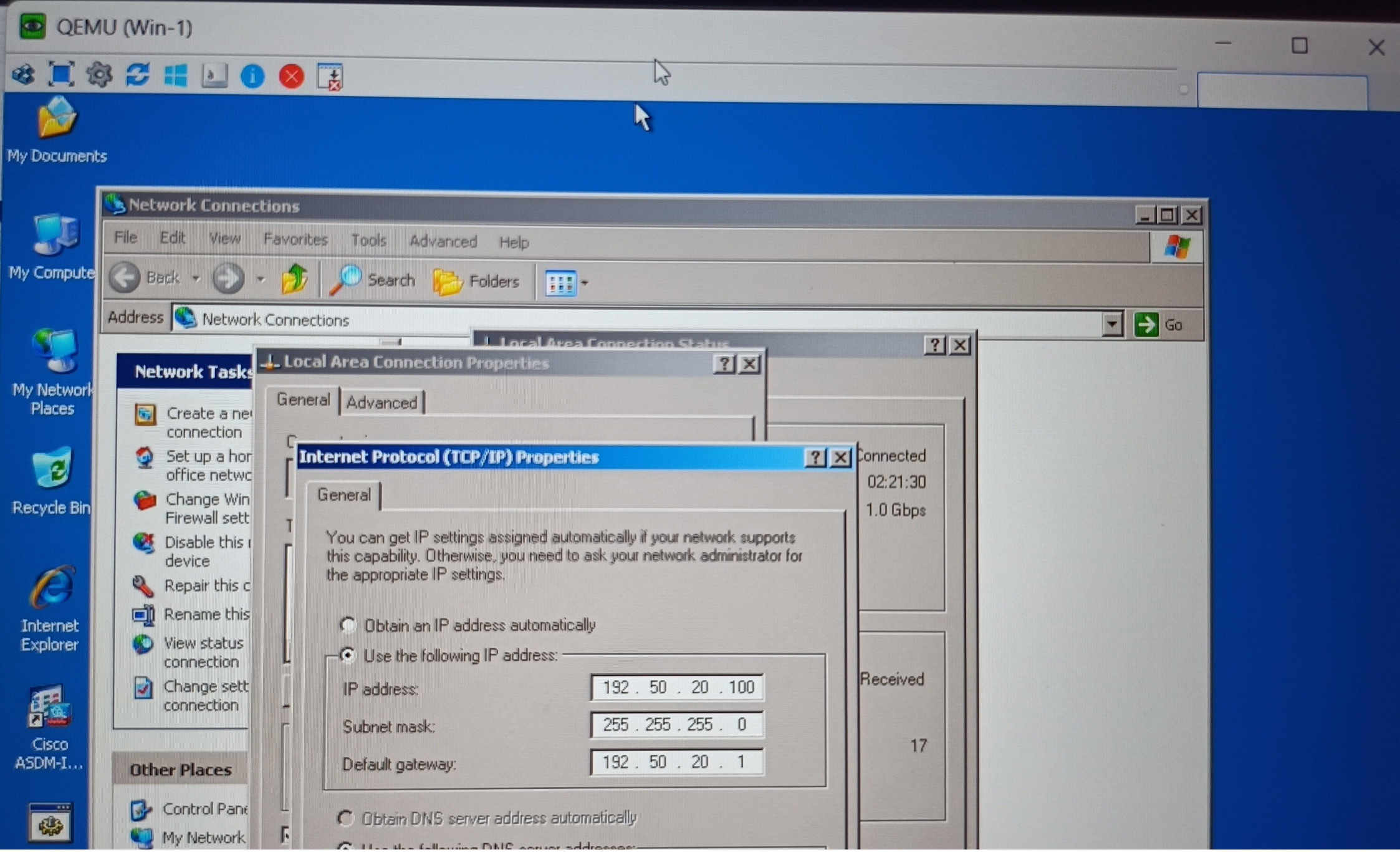
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from Pfr

Gateway of last resort is not set

```
S    12.0.0.0/8 [1/0] via 26.0.0.1
S    23.0.0.0/8 [1/0] via 26.0.0.1
S    24.0.0.0/8 [1/0] via 26.0.0.1
    26.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        26.0.0.0/8 is directly connected, Ethernet0/1
L        26.0.0.2/32 is directly connected, Ethernet0/1
S    32.0.0.0/8 [1/0] via 26.0.0.1
S    45.0.0.0/8 [1/0] via 26.0.0.1
S    172.36.0.0/16 [1/0] via 26.0.0.1
S    192.50.20.0/24 [1/0] via 26.0.0.1
    223.0.50.0/24 is variably subnetted, 2 subnets, 2 masks
C        223.0.50.0/24 is directly connected, Ethernet0/0
L        223.0.50.1/32 is directly connected, Ethernet0/0
```

R6#

R6#





My Documents



My Computer

My Network
Places

Recycle Bin

Internet
ExplorerCisco
ASDM-I...

Old Java ASD

Network Connections

File Edit View Favorites Tools Advanced Help

Back Forward Up Search Folders

Address Network Connections Go

Network Tasks

- Create a new connection
- Set up a home office network
- Change Windows Firewall settings
- Disable this connection
- Repair this connection
- Rename this connection
- View status of connection
- Change settings of connection

Other Places

- Control Panel
- My Network Places
- My Documents
- My Computer

Local Area Connection Properties

General Advanced

Internet Protocol (TCP/IP) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address: 223 . 0 . 50 . 10

Subnet mask: 255 . 255 . 255 . 0

Default gateway: 223 . 0 . 50 . 1

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server: . . .

Alternate DNS server: . . .

Connected
02:20:10
1.0 Gbps

Received
16

Close



My Documents

My Computer

My Network Places

Recycle Bin

Internet Explorer

Cisco ASDM-1...

Old Java ASD



Network Connections

File Edit View Favorites Tools Advanced Help

Back Forward Stop Search Folders

Address Network Connections

Network Tasks

- Create a new connection
- Set up a home or office network
- Change Windows Firewall settings
- Disable this network device
- Repair this connection
- Rename this connection
- View status of connection
- Change settings of connection

Other Places

- Control Panel
- My Network Places
- My Documents
- My Computer

Details

Local Area Connection Properties

General Advanced

Internet Protocol (TCP/IP) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address: 172 . 36 . 0 . 10

Subnet mask: 255 . 255 . 0 . 0

Default gateway: 172 . 36 . 0 . 1

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server: . . .

Alternate DNS server: . . .

Connected

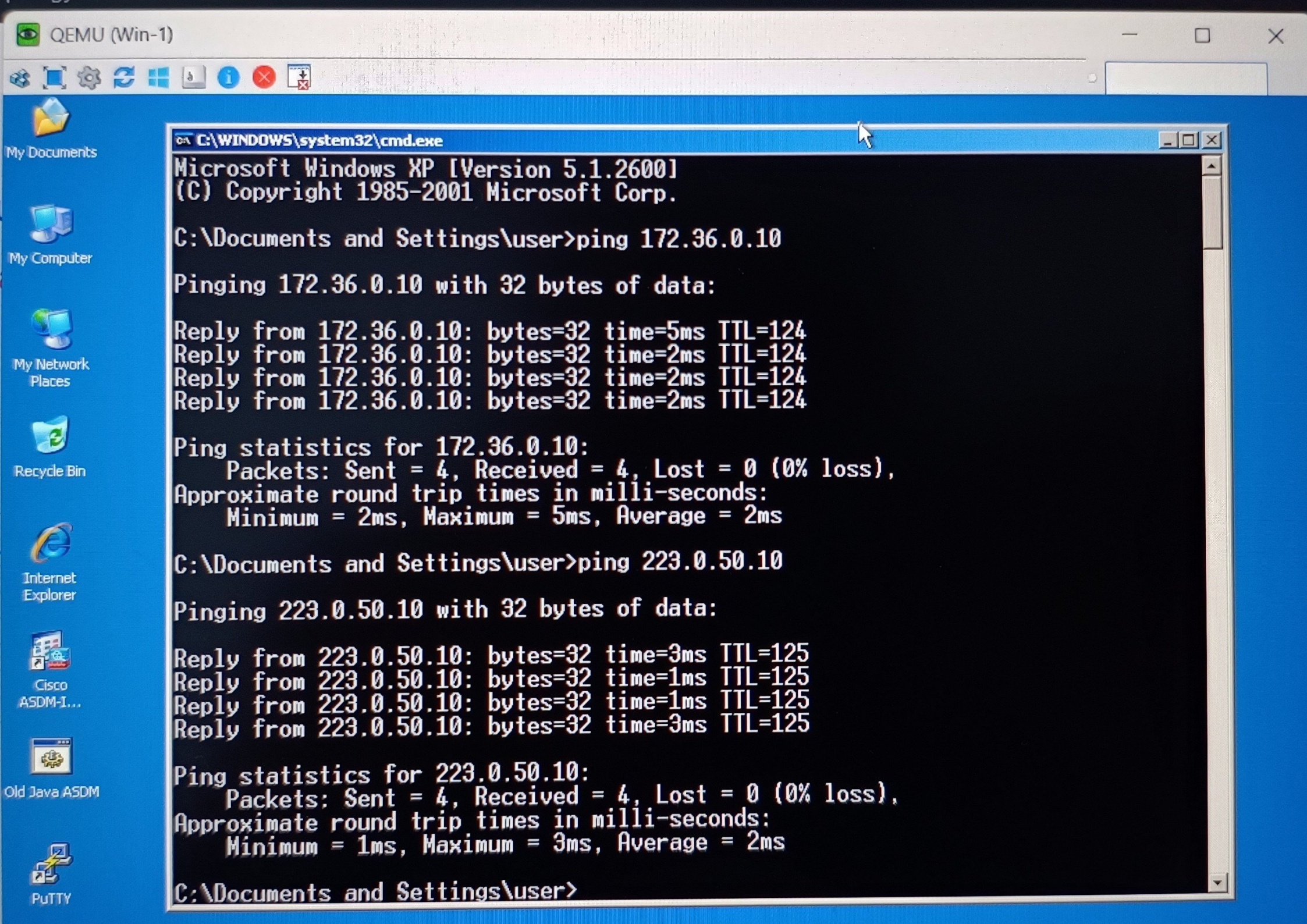
02:20:34

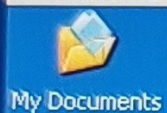
1.0 Gbps

Received

16

Close





C:\WINDOWS\system32\cmd.exe

Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ping 192.50.20.100

Pinging 192.50.20.100 with 32 bytes of data:

Reply from 192.50.20.100: bytes=32 time=1ms TTL=125
Reply from 192.50.20.100: bytes=32 time=2ms TTL=125
Reply from 192.50.20.100: bytes=32 time=2ms TTL=125
Reply from 192.50.20.100: bytes=32 time=2ms TTL=125

Ping statistics for 192.50.20.100:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Documents and Settings\user>ping 172.36.0.10

Pinging 172.36.0.10 with 32 bytes of data:

Reply from 172.36.0.10: bytes=32 time=2ms TTL=124
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 172.36.0.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 2ms, Average = 2ms

C:\Documents and Settings\user>



My Documents

My Computer

My Network Places

Recycle Bin

Internet Explorer

Cisco ASDM-1...

Old Java ASDM

PuTTY

C:\WINDOWS\system32\cmd.exe

Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ping 192.50.20.100

Pinging 192.50.20.100 with 32 bytes of data:

Reply from 192.50.20.100: bytes=32 time=2ms TTL=124
Reply from 192.50.20.100: bytes=32 time=7ms TTL=124
Reply from 192.50.20.100: bytes=32 time=2ms TTL=124
Reply from 192.50.20.100: bytes=32 time=3ms TTL=124

Ping statistics for 192.50.20.100:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 7ms, Average = 3ms

C:\Documents and Settings\user>ping 223.0.50.10

Pinging 223.0.50.10 with 32 bytes of data:

Reply from 223.0.50.10: bytes=32 time=3ms TTL=124
Reply from 223.0.50.10: bytes=32 time=2ms TTL=124
Reply from 223.0.50.10: bytes=32 time=2ms TTL=124
Reply from 223.0.50.10: bytes=32 time=2ms TTL=124

Ping statistics for 223.0.50.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>_